

IPCC Tier 2 Livestock GHG Uncertainty Calculator

Beta Testing, How to Review the Tool and Send Feedback

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Help us improve the tool: a 20-minute review

Thank you for agreeing to review the **IPCC Tier 2 Livestock GHG Uncertainty Calculator**. This short guide walks you through a quick hands-on test using a small practice dataset, then shows you how to record and send your feedback. Plan for about **20–30 minutes**. You don't need to be an R user or install anything: the tool runs in your web browser.

What you'll need

Three files (attached to the invitation email, or downloadable from the app's **Resources** tab):

1. **beta_sample_inventory.xlsx**: a small practice dataset to load into the tool.
2. **beta_feedback_template.xlsx**: the form where you write your feedback.
3. **This guide**.

Step 1: Open the tool

Go to <https://mlolita26.shinyapps.io/cattle-ghg-uncertainty/>. The calculator opens straight away, **no login is needed**. (Only the optional *AI Translator* on the Resources tab asks for an email; you can ignore it for this review.)

Step 2: Load the practice dataset (*about 5 min*)

The practice file is a small, made-up inventory: one anonymised “**Country**” with **two production systems** (Commercial dairy and Extensive beef), each split into **two animal sub-categories**. It's deliberately tiny so you can see how every part of the tool behaves without wading through a real national dataset.

1. Open the **Data Input** tab.
2. Under **Upload**, select **beta_sample_inventory.xlsx**.
3. Tick the **emission sources** to include: start with all of them (enteric CH₄, manure CH₄, direct and indirect N₂O).

(The built-in “Country X / Country Y” examples in the dropdown are also there to play with, but please base your review on the practice file so we're all looking at the same thing.)

Step 3: Walk through the tabs (*about 10 min*)

Go through these in order, and jot down anything that is **confusing, wrong, or missing**:

- **QA/QC**: the automatic data checks and any flags they raise.
- **Uncertainty**: the $\pm$ ranges shown for each parameter.
- **Correlations**: how parameters move together (the practice file includes multi-year data, so this tab has something to display).
- **Simulate**: run the Monte Carlo simulation; look at the totals and the distribution chart.
- **Sensitivity**: which inputs drive most of the uncertainty (the tornado chart).
- **IPCC Report**: the summary tables and the downloadable Word report.

If you have a real inventory of your own, please try uploading that too and note how it goes.

Step 4: Record your feedback (*the main step*)

Open **beta_feedback_template.xlsx** and add **one row per observation** in the **Feedback** sheet. Use the drop-down menus in each cell:

- **Tab**: where you saw it (Home, Data Input, QA/QC, . . . , Contact).
- **Type** and **Severity**: see the tables below.
- **Description**: what you observed and what you expected instead.
- **Suggested fix / comment**: optional; your idea for the right behaviour.

Leave the **Status** column blank: the development team fills that in as items are addressed.

Type	Use it when...
Bug	The tool crashes, produces wrong numbers, or behaves unexpectedly
Correction	Something is factually wrong: a method, equation, IPCC reference, or number
Enhancement	A feature or change that would make the tool more useful or easier to use
General	A general comment, question, or observation

Severity	Meaning
Critical	Must fix before release: a calculation error or crash
High	Should fix before release: a significant accuracy or usability issue
Medium	Can fix in the next iteration: a minor issue that doesn't block use
Low	Nice to have: cosmetic or very minor

Severity

Meaning

Step 5: Send it back

Save the completed `beta_feedback_template.xlsx` and **email it to M.Lolita@cgiar.org**. If several people are reviewing, each can send their own file: the *Reviewer name* column keeps everyone's comments distinct.

Two things you might notice (and can ignore)

- The green “**Feedback**” button at the bottom-right of the app is for quick, one-off bug reports once the tool is in wider use. For this structured review, please use the **Excel form** above instead.
- `questionnaire.md` / `questionnaire.docx` (in the Resources area) is a data-preparation form for the *AI Translator*: it is **not** this feedback form. You can ignore it for the review.

What makes feedback most useful

- **Be specific**: name the tab and the exact button, table, chart, or text.
- **Say what you expected** versus what actually happened.
- **One observation per row**, so each item can be tracked and resolved individually.

Thank you: your review directly shapes what we fix before the tool is released more widely.